



Machine Vision Course
Prof. Hamidreza Pourreza
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MiniProject-6

Classic Digit Classifier via Fourier Descriptors

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Project Abstract

This project implements a lightweight, single-stage classifier for Persian digit recognition using Fourier Descriptors (FD) and k-Nearest Neighbors (kNN). From a single page image containing 8 rows of digits (each row: 9→0), we extract the external contour of each digit, resample it uniformly along arc-length ($N = 512$), and compute complex FD via the DFT. To achieve invariance, we remove translation by zeroing the DC term, normalize scale by dividing by $|C_1|$, and enforce rotation/start invariance via phase-locking to C_1 . The feature vector is formed by concatenating the real and imaginary parts of the lowest-frequency harmonics; we evaluate dimensionality reductions that retain 20%, 10%, 5%, 2%, and 1% of the spectrum. Classification uses distance-weighted kNN ($k = 5$). Performance is measured with a Leave-One-Row-Out protocol (each row held out in turn). Results show stable accuracy of **0.963** for 20%, 10%, 5%, and 2% retention, with a modest drop to **0.950** at 1%. Confusions primarily occur between visually similar shapes (e.g., 2↔3; occasional 7→2). The approach is simple, fast, and robust to geometric variations, making it suitable for resource-constrained OCR pipelines where clean contours are available.

Keywords: Fourier Descriptors (FD), k-Nearest Neighbors (kNN), Persian Digit Recognition (Shape-based OCR)

Algorithm Pipeline Description

1) Preprocess → External Components

- **Gray → Binary:** Otsu (inverted) so digits= 1, background= 0. Light open/close if needed (avoid strong closing).
- **Contours:** Extract **external** contours (RETR_EXTERNAL, CHAIN_APPROX_NONE). Keep (x, y, w, h) , $y_{\text{top}} = y$, $x_{\text{left}} = x$ per component.

2) Row Building (robust)

- **Adaptive gaps (primary):** sort by y_{top} . Start a row; add next component if $|y_{\text{top}} - \bar{y}_{\text{row}}| \leq \text{tol}$, where $\text{tol} = \max(\text{row_gap_min}, \text{row_gap_tol_rel} \times \text{median_row_height})$.
- **Fallback K-Means:** if rows $\neq 8$, cluster y -centers with $K = 8$, $n_{\text{init}} = 10$, $\text{random_state} = 0$. Order rows by y , items by x .

3) Per-Row Cleanup & Selection

- **Drop rightmost bullet (if present):** last-by- x is bullet if $\text{area}_{\text{last}} < \text{bullet_area_rel} \times \text{median_area}(\text{row})$ and circularity $4\pi A/P^2 > \text{bullet_circularity_min}$.
- **Filter noise:** keep components with $\text{area} \geq \text{min_component_area_rel} \times \text{median_area}(\text{row})$ and $\text{height} \geq \text{min_component_height_rel} \times \text{median_height}(\text{row})$.
- **Split merged (if < 10):** choose widest-by (width $\times$ area). In its ROI:
 - 1) column-sum profile → smooth (win= 5),
 - 2) ignore margins $\approx 0.08 \times w$, find global valley,
 - 3) zero a thin vertical band ($\approx 0.01 \times w$) → re-extract externals,
 - 4) replace the wide component with the two parts; repeat (guarded).
- **Keep leftmost 10**, label 9→0 left→right.

Default thresholds (good starting point):

$\text{bullet_area_rel}=0.18$, $\text{bullet_circularity_min}=0.85$,
 $\text{min_component_area_rel}=0.03$, $\text{min_component_height_rel}=0.35$,
 $\text{row_gap_tol_rel}=0.45$, $\text{row_gap_min}=6$.

4) Uniform Contour Resampling

- Close loop if needed. Compute cumulative arc-length; **resample to** $N = 512$ equally spaced points.
- Map to complex: $z_n = x_n + i y_n$, $n = 0..N - 1$.

Algorithm Pipeline Description

5) Fourier Descriptors with Invariances

- **Center (translation):** $\tilde{z}_n = z_n - \frac{1}{N} \sum z \Rightarrow C_0 = 0$.
- **FFT:** $C_k = \frac{1}{N} \sum \tilde{z}_n e^{-i2\pi kn/N}$.
- **Scale:** $\hat{C}_k = C_k / |C_1|$.
- **Rotation & start:** phase-lock all coefficients by $-\arg(C_1)$: $\check{C}_k = \hat{C}_k e^{-i \arg(C_1)}$.

Why this set: center removes translation, $/|C_1|$ removes size, **phase-lock** removes global rotation & starting index—leaving shape.

6) Feature Vector (per retention $p\%$)

- Drop DC; keep lowest $M = \max(1, \lfloor p \cdot (N - 1) \rfloor)$ harmonics: $\check{C}_1 \dots \check{C}_M$.
- **Real features:** $[\Re(\check{C}_{1..M}), \Im(\check{C}_{1..M})] \in \mathbb{R}^{2M}$.
- **Optional mid-band emphasis:** multiply $k \in [4, 12]$ by ~ 1.4 (applied to both Re/Im).
- For $N = 512$: $N - 1 = 511 \rightarrow 20\%$: $M = 102 \rightarrow \mathbf{dim=204}$ · 10% : $M = 51 \rightarrow \mathbf{102}$ · 5% : 50 · 2% : 20 · 1% : 10 .

(Optional: z-score features using **train** mean/std; usually not necessary here.)

7) Classifier & Evaluation

- **Classifier:** kNN, $k = 5$, weights="distance", Euclidean.
- **Protocol (LORO):** For each row r : train on other 7 rows, test on row r ; concatenate predictions.
- **Report per $p\%$:** overall accuracy; 10×10 confusion matrix; (optional) per-class metrics.
- **Artifacts:**
 - Overlays with predicted digits on held-out rows,
 - CM heatmaps,
 - accuracy.txt, confusion.csv/txt, classification_report.txt,
 - config.txt capturing run settings.

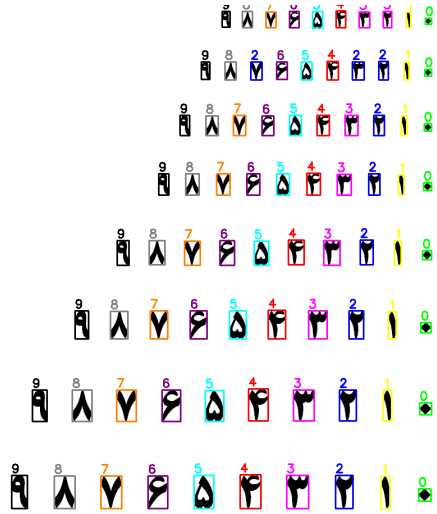
Results and Discussion

- **Accuracy vs. retention.** Performance is flat at **0.9625** for 2–20% FD retention, then dips to **0.9500** at 1%. This plateau means the lowest harmonics already encode the class-discriminative outline; adding more bands above $\sim 2\%$ mostly restores fine detail without changing decisions. The 1% setting removes just enough curvature/indentation cues to cause occasional slips.
- **Per-class errors.** From 2–20%, the only consistent misses are single confusions for **2, 3, and 7** (recall 0.88 each), while all other digits are perfectly recalled. At 1%, **0** also drops to recall 0.88. These four are exactly the ones whose silhouettes can become deceptively similar when over-smoothed (e.g., $2 \leftrightarrow 3$'s open curves; 7's slanted stroke rounding toward curved classes.)
- **Interpretation.** Low-frequency FD captures global geometry (overall roundness, main bends, aspect) which is sufficient for this dataset. When truncation is too aggressive (1%), the boundary “loses character”: distinctive corners and local bends blur, increasing overlap in FD space for look-alike pairs—most notably (2,3).

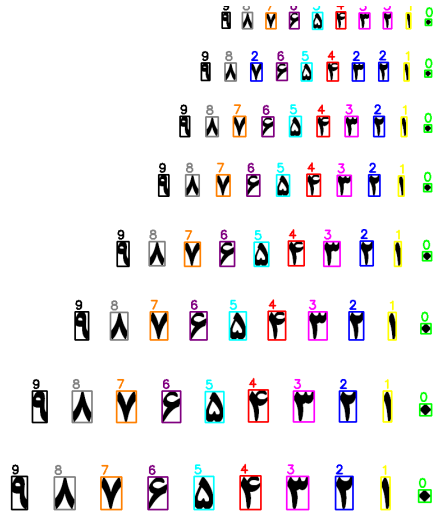
Results & Visual Analysis

These figures summarize performance across FD retention levels and visualize per-row predictions under the LORO protocol. The confusion matrices (keep 1%, 2%, 5%, 10%, 20%) show a near-saturated diagonal from 2–20%, indicating stable accuracy with only a few residual confusions—primarily $2 \leftrightarrow 3$ and $7 \rightarrow 2$ —while the 1% matrix introduces one extra error ($0 \rightarrow 6$) consistent with over-smoothed contours at extreme compression. The accompanying LORO overlays corroborate the matrices: most boxes are correctly labeled across all rows, and the rare mistakes occur exactly in the classes highlighted by the off-diagonal cells. Together, the plots demonstrate that low-frequency Fourier descriptors already carry the discriminative shape information; adding more harmonics above $\sim 2\%$ adds visual detail but does not change decisions, while pushing down to 1% removes just enough curvature cues to create a few look-alike confusions.

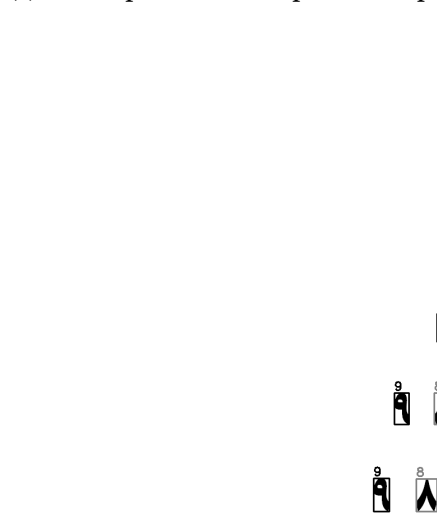
Results & Visual Analysis



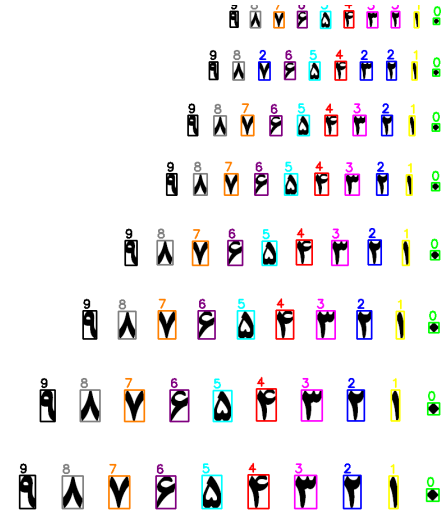
(a) LORO predictions, 20 percent kept



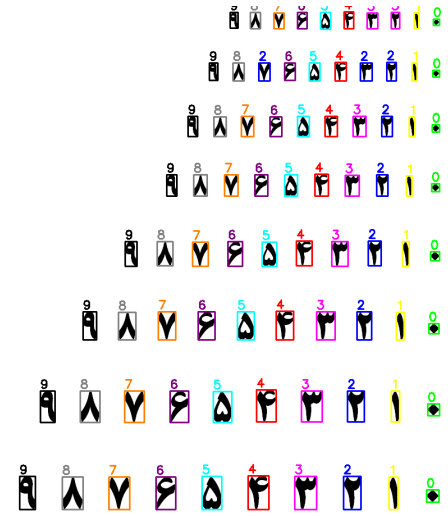
(c) LORO predictions, 5 percent kept



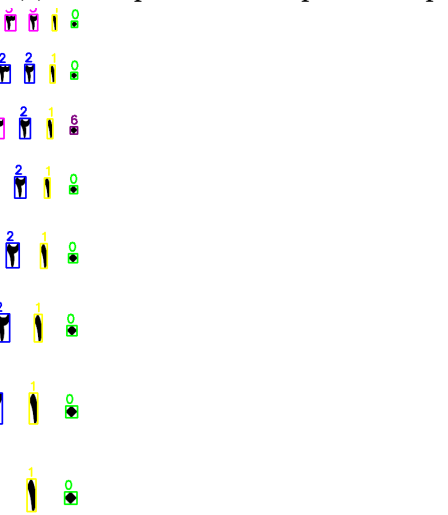
(e) LORO predictions, 1 percent kept



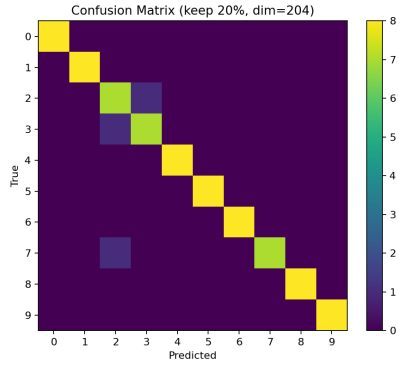
(b) LORO predictions, 10 percent kept



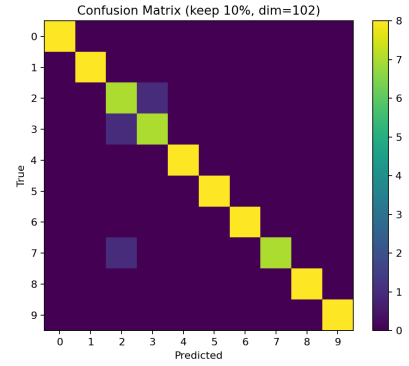
(d) LORO predictions, 2 percent kept



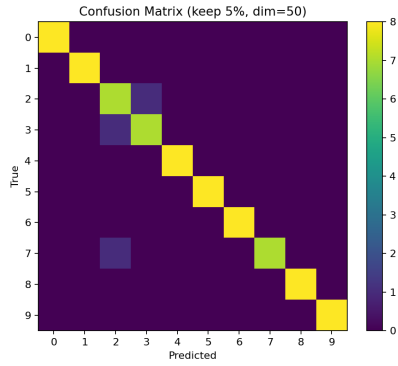
Results & Visual Analysis



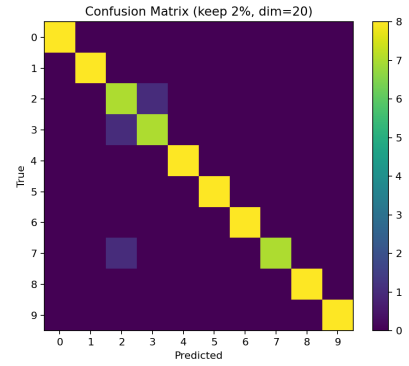
(a) Confusion matrix, 20 percent kept



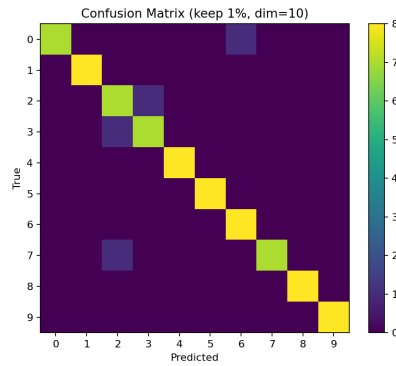
(b) Confusion matrix, 10 percent kept



(c) Confusion matrix, 5 percent kept



(d) Confusion matrix, 2 percent kept



(e) Confusion matrix, 1 percent kept